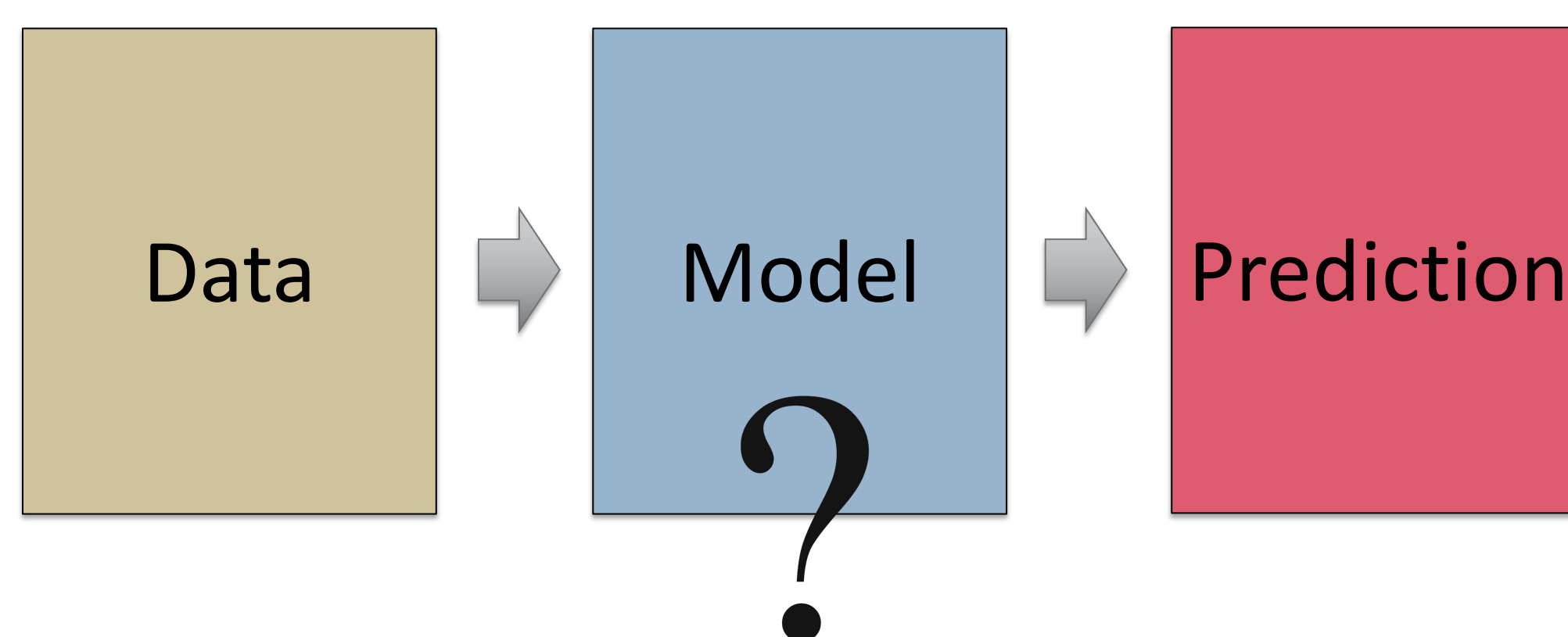


DISENTANGLING INFLUENCE: USING DISENTANGLED REPRESENTATIONS TO AUDIT MODEL PREDICTIONS

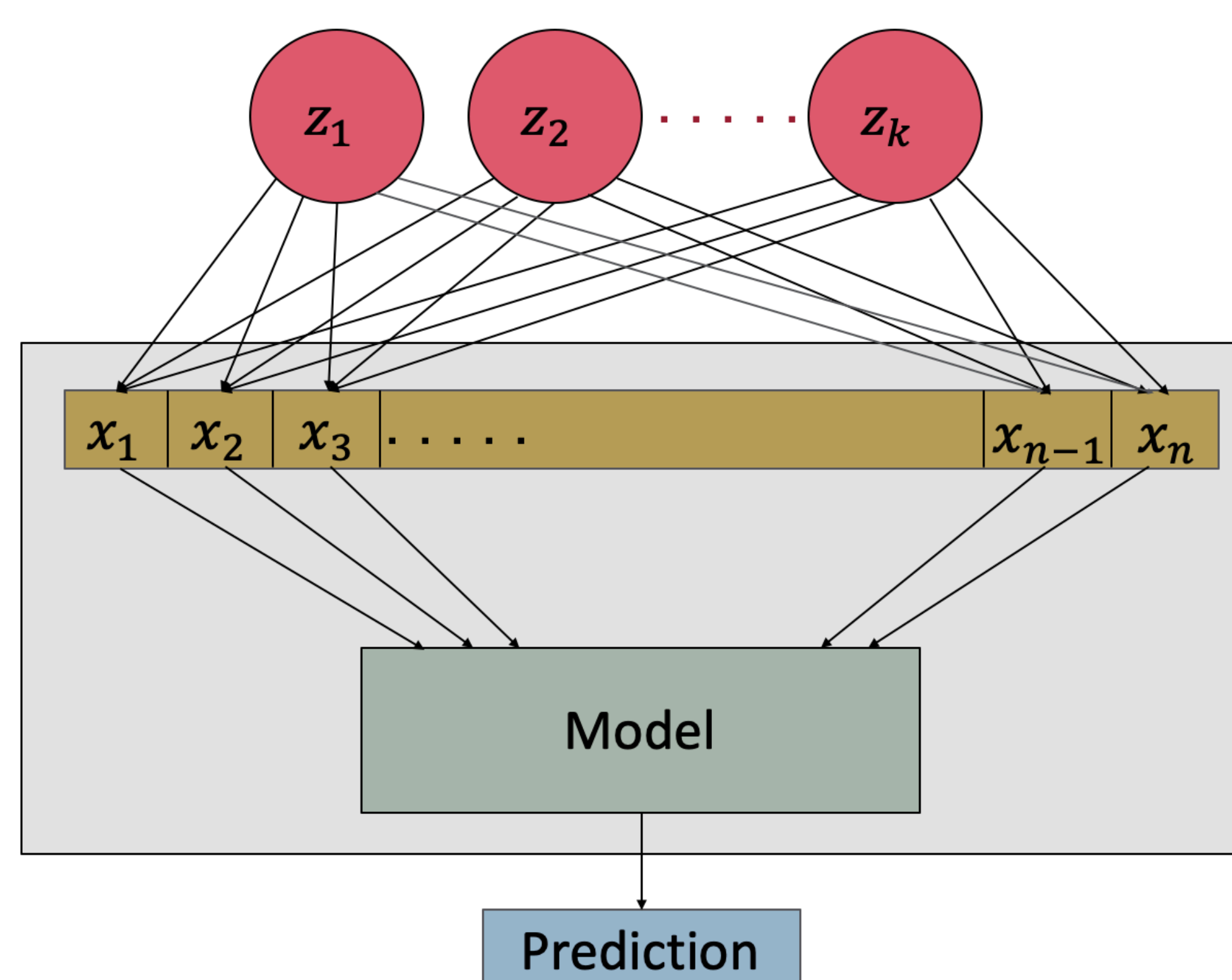
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How do Machines Understand Data?



Explainability is important for many aspects of fairness in machine learning including transparency and contestability. Direct feature importance methods cannot measure when a feature influences a model prediction via a *proxy* variable — a feature used by the model which is highly related to a feature of interest.

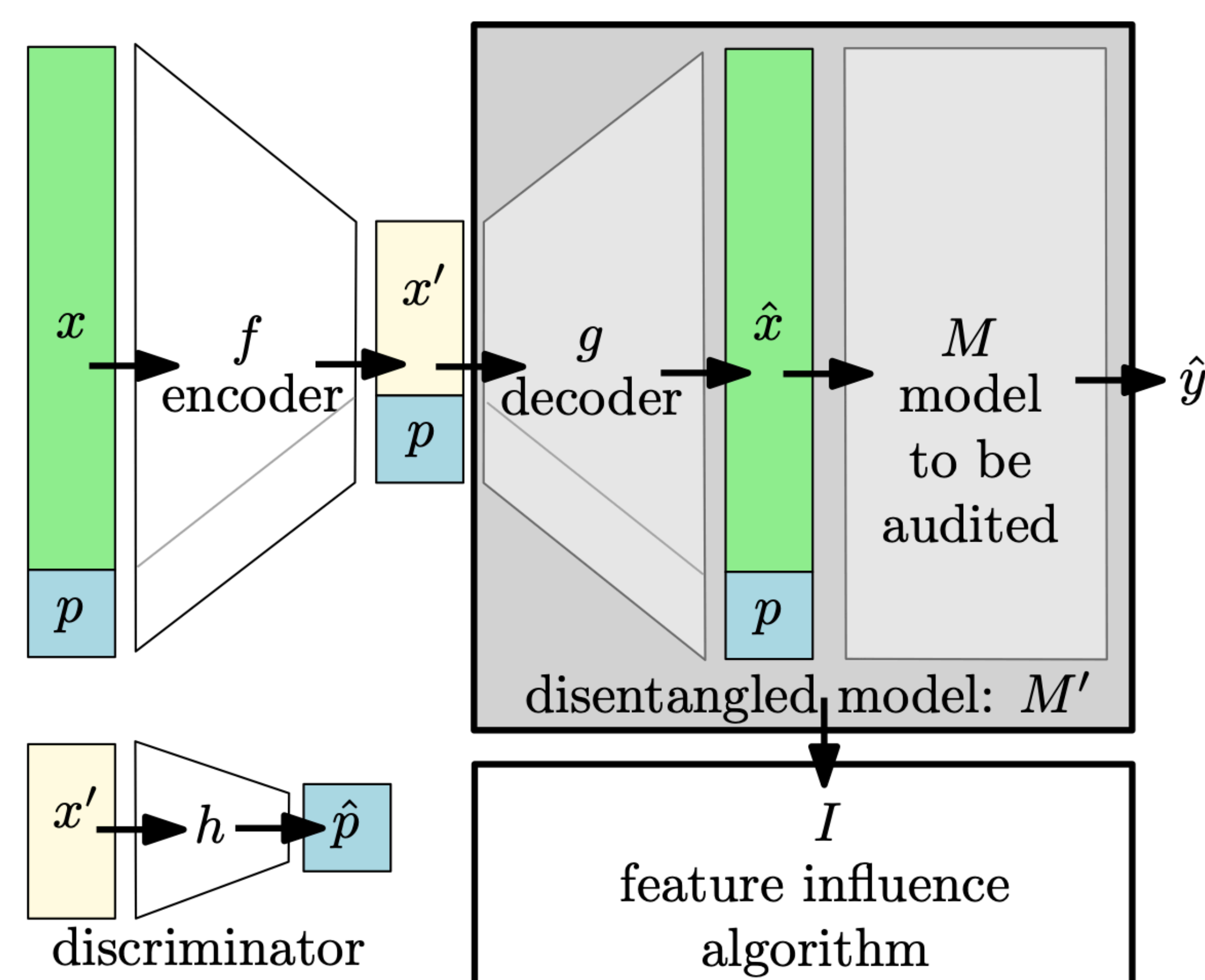
Disentangled Representations



Independent factors of variation z_i generate variables x_i from which a model makes a prediction. We use ideas from disentangled representations to compute the indirect influence of each latent factor z_i .

Indirect Influence: Detecting when a feature affects a model prediction through a *proxy* variable.

Model Architecture



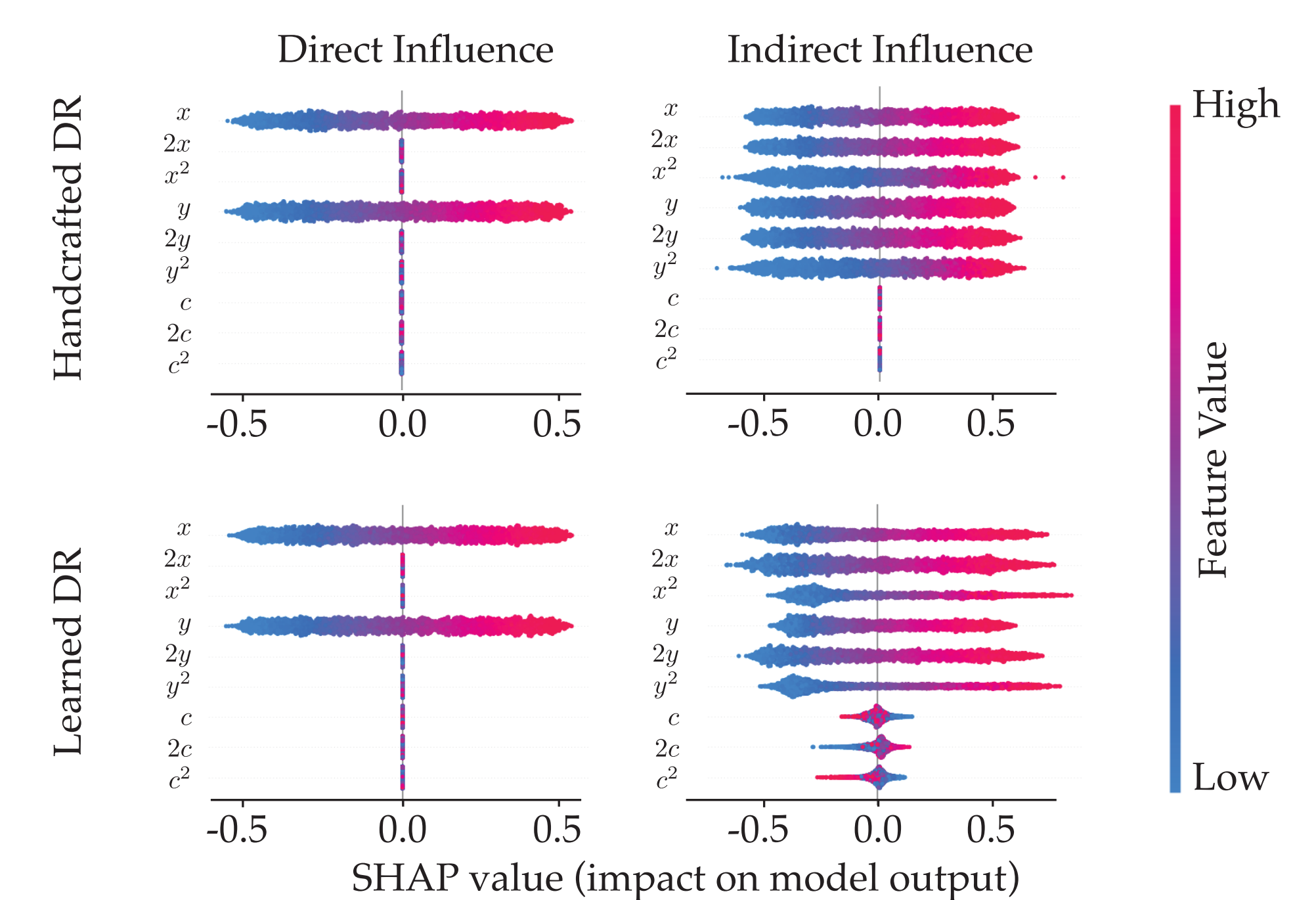
Our Proposed Definition of Indirect Influence:

$$II_p(M, (p, x)) := DI_p(M \circ g, (p, x'))$$

We reduce the indirect influence problem to the direct influence problem, for which there are existing solutions. We learn each feature in x as a function of independent factors of variation p and x' in order to determine how each feature in x acts as a proxy for p .

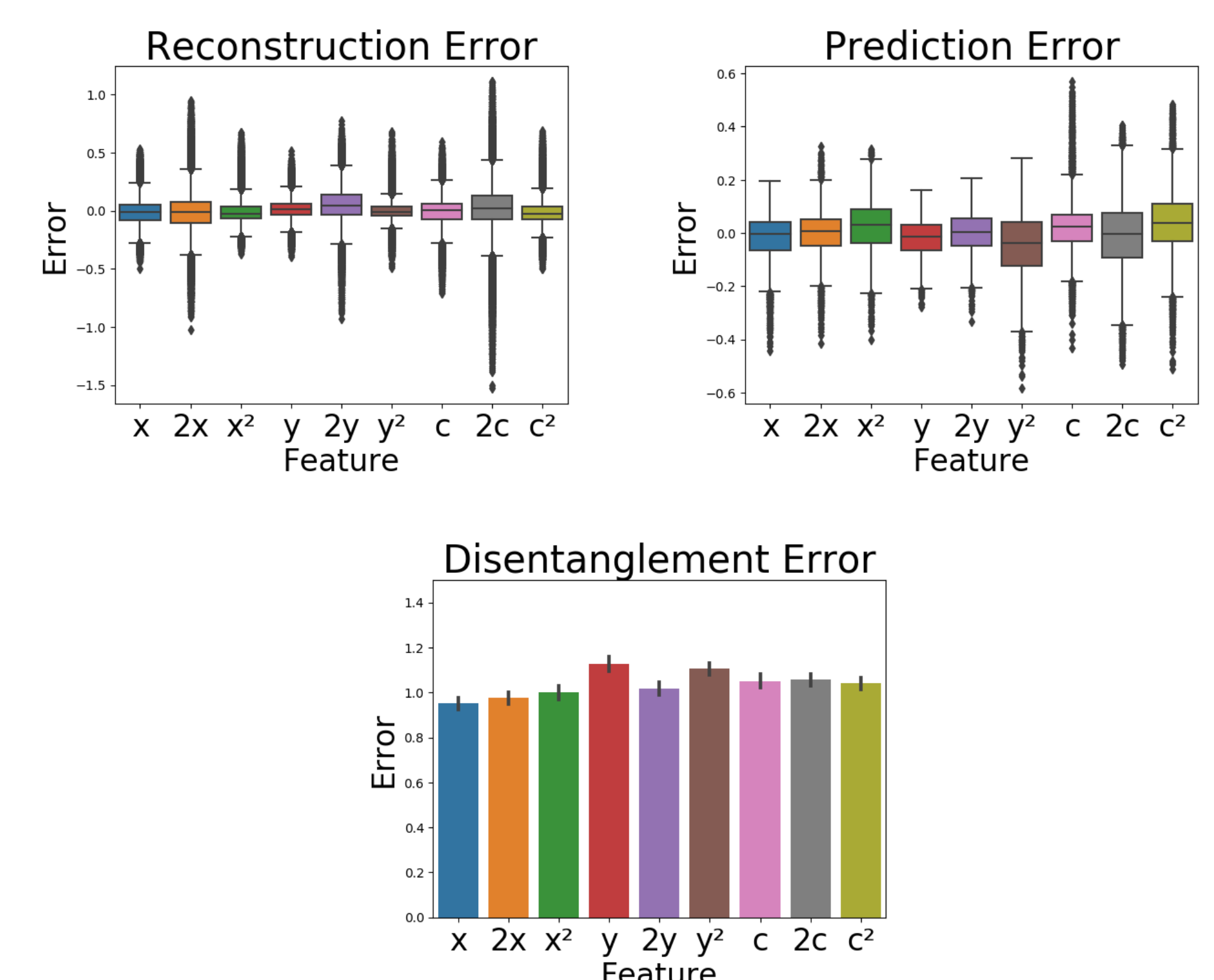
Synthetic Experiments

Latent Factors: $x, y, c \sim i.i.d. \text{Unif}(0,1)$
Features: $x, 2x, x^2, y, 2y, y^2, c, 2c, c^2$
Label: $x + y$



The direct influence method SHAP (left) reports that the only features used by the model are x and y . After wrapping the model using our method to generate a disentangled model, the same SHAP method is used to compute indirect influence (right), where we can see that $2x, 2y, x^2, y^2$ all act as proxies for x and y .

Error Metrics



Code: <https://github.com/charliemarx/disentangling-influence>
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